

(c) Calculate the work done to bring a test charge +2 μC from infinity to the point Q. Assume the charge +9 μC is held fixed at origin and +2 μC is brought from infinity to P.

Solution

(a) Electric potential at point P is given by

$$V_P = \frac{1}{4\pi\epsilon_\circ}\frac{q}{r_P} = \frac{9\times10^9\times9\times10^{-6}}{10} = 8.1\times10^3\ \text{V}$$

Electric potential at point Q is given by

$$V_Q = \frac{1}{4\pi\epsilon_\circ}\frac{q}{r_Q} = \frac{9\times10^9\times9\times10^{-6}}{16} = 5.06\times10^3\ \text{V}$$

Note that the electric potential at point Q is less than the electric potential at point P. If we put a positive charge at P, it moves from P to Q. However if we place a negative charge at P it will move towards the charge +9 μC.

The potential difference between the points P and Q is given by

$$\Delta V = V_P - V_Q = +3.04\times10^3\ \text{V}$$

(b) Suppose we replace the charge +9 μC by −9 μC, then the corresponding potentials at the points P and Q are,

$$V_P = -8.1\times10^3\ \text{V},\ V_Q = -5.06\times10^3\ \text{V}$$

Note that in this case electric potential at the point Q is higher than at point P.

The potential difference between the points P and Q is given by

$$\Delta V = V_P - V_Q = -3.04\times10^3\ \text{V}$$

(c) The electric potential V at a point Q due to some charge is defined as the work done by an external force to bring a unit positive charge from infinity to Q. So to bring the q amount of charge from infinity to the point Q, work done is given as follows.

$$W = qV$$

$$W_Q = 2\times10^{-6}\times5.06\times10^3 = 10.12\times10^{-3}\ \text{J}.$$

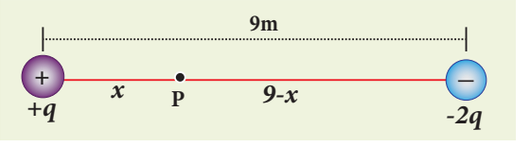
EXAMPLE 1.13

Consider a point charge $+q$ placed at the origin and another point charge $-2q$ placed at a distance of 9 m from the charge $+q$. Determine the point between the two charges at which electric potential is zero.

Solution

According to the superposition principle, the total electric potential at a point is equal to the sum of the potentials due to each charge at that point.

Consider the point at which the total potential zero is located at a distance x from the charge $+q$ as shown in the figure.



Since the total electric potential at P is zero,

$$V_{tot} = \frac{1}{4\pi\epsilon_\circ}\left(\frac{q}{x} - \frac{2q}{(9-x)}\right) = 0 \ \text{(or)}$$

$$\frac{q}{x} = \frac{2q}{(9-x)} \ \text{(or)}$$

$$\frac{1}{x} = \frac{2}{(9-x)}$$

Hence, x = 3 m

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